UNITED STATES OF AMERICA

BEFORE THE

FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	))	**Docket No. RM26-4-000**

COMMENTS OF
SOUTHWEST POWER POOL INC.

Southwest Power Pool, Inc. ("SPP") respectfully submits the following comments in response to the Federal Energy Regulatory Commission's ("Commission") Notice Inviting Comments[1] related to the Secretary of Energy's proposed Advance Notice of Proposed Rulemaking for consideration and final action by the Commission.[2]

I. INTRODUCTION

As the Secretary of Energy points out in the Proposed ANOPR, the United States is experiencing an increasing demand for electricity growth due, in large part, to the rapid growth of large loads. SPP commends the Department of Energy and the Commission for initiating the Proposed ANOPR and providing an opportunity to comment on the possible reforms needed to address the interconnection of large loads. Any standardized interconnection rules adopted by the Commission will have significant impacts on SPP's

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1 *Interconnection of Large Loads to the Interstate Transmission System*, Notice Inviting Comments, Docket No. RM26-4-000 (Oct. 27, 2025) ("Notice").

2 *Interconnection of Large Loads to the Interstate Transmission System*, Advance Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 27, 2025) ("Proposed ANOPR").

load interconnection processes, transmission service processes, generator interconnection processes, and the SPP Integrated Marketplace.

SPP encourages the Commission to provide flexibility in any rules adopted based on the vast regional differences across the country. Rather than pursuing a "one-size fits all" approach, the Commission should consider allowing each Regional Transmission Organization ("RTO") and Independent System Operator ("ISO") ("RTO/ISO") and its stakeholders the latitude to design for these differences. For example: (1) differences in states with vertically integrated utilities that have the obligation to serve versus deregulated states; (2) differences in RTOs/ISOs that are located within one state versus located across multiple states with varying state laws pertaining to retail sales of electricity, retail wheeling of electricity, generation siting, load and generation co-ownership, or load and generation operation; (3) differences in regions of the country that do not participate in an RTO/ISO versus those that do participate; and (4) differences that result from whether the load is interconnecting in the Eastern Interconnection, Western Interconnection, or the Texas Interconnection.

SPP is at the forefront of the development of solutions to interconnect large load, as discussed in more detail below. During this process, SPP urges that the Commission provide regional deference to any large load interconnection proposals specifically designed to address the needs of SPP and its stakeholders.

II. SPP'S CURRENT ACTIVITY RELATING TO LOAD CONNECTION

SPP, like other RTOs, ISOs, and utilities, is facing significant operational, planning, and market challenges in response to unprecedented growth in large, often non-traditional, load interconnection requests. Consistent with its responsibilities, SPP is rising to the challenge of integrating these large loads into the RTO footprint in a manner that facilitates the continued safe and reliable operation of the Transmission System.[3] SPP is addressing the challenge of interconnecting large loads in a multi-step approach beginning with filings made in 2025 and planned filings in 2026.

Under SPP's Open Access Transmission Tariff ("Tariff"),[4] Transmission Customers[5] are required to have sufficient Designated Resources[6] to serve their existing load and any proposed interconnection of new load. Because of the substantial increase in requests for load additions, a significant portion of which are large load Interconnection Requests,[7] current Transmission Customers are unable to request studies of new load

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3 Tariff at Part I, Definitions T defines Transmission System as "the facilities used by the Transmission Provider to provide transmission service under Part II, Part III and Part IV of the Tariff."

4 Southwest Power Pool, Inc., Open Access Transmission Tariff, Sixth Revised Volume No. 1 ("Tariff"). References in these comments to "Tariff" refer to the version of SPP's Tariff currently in effect. All capitalized terms not otherwise defined in these comments shall have the definitions assigned by the Tariff.

5 Tariff at Part I, Definitions T defines Transmission Customer as "any Eligible Customer (or its Designated Agent) that (i) executes a Service Agreement, or (ii)requests in writing that the Transmission Provider file with the Commission, a proposed unexecuted Service Agreement to receive transmission service under Part II of the Tariff. This term is used in the Part I Common Service Provisions to include customers receiving transmission service under Part II and Part III of this Tariff."

6 Tariff at Part I, Definitions D defines Designated Resource as "any designated generation resource owned, purchased or leased by a Transmission Customer to serve load in the SPP Region. Designated Resources do not include any resource, or any portion thereof, that is committed for sale to third parties or otherwise cannot be called upon to meet the Transmission Customer's load on a non-interruptible basis."

7 Tariff at Attachment V, Section 1 defines Interconnection Request as "an Interconnection Customer's request, in the form of Appendix 3 or Appendix 5 to this GIP, in accordance with the Tariff, to interconnect a new Generating Facility,

additions because they do not have sufficient Designated Resources to serve the load. To remedy this issue, SPP proposed revisions to its Tariff in Docket No. ER25-2430 to implement a Provisional Load Process[8] to allow Transmission Customers to request studies of new load Interconnection Requests, provided that the Transmission Customer has planned generation to serve the load additions. The Commission approved these proposed revisions on October 10, 2025.[9]

Recently, in Docket No. ER26-247,[10] SPP proposed further revisions to the Tariff to account for these large load additions. In this filing, SPP proposed (1) a new category of load, termed a High Impact Large Load ("HILL");[11] (2) enhanced study requirements for HILLs to enable SPP to assess the load's reliability impacts on the Transmission System;

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to interconnect a Replacement Generating Facility, to increase the capacity of, or make a Material Modification to the operating characteristics of, an Existing Generating Facility that is interconnected with the Transmission System."

8 Tariff at Part I, Definitions P defines Provisional Load Process as "this process, as described in Attachment AX of the Tariff, allows the Transmission Provider to assess the request for delivery point additions or changes when the Transmission Customer does not have enough existing Designated Resources to serve the increased load plus losses."

9 *Sw. Power Pool Inc.*, 193 FERC ¶ 61,018 (Oct. 10, 2025).

10 Submission of Tariff Revisions to Add the High Impact Large Load Processes and High Impact Large Load Generation Assessment of Southwest Power Pool, Inc., Docket No. ER26-247-000 (Oct. 24, 2025) ("SPP's High Impact Large Load Filing").

11 SPP's High Impact Large Load Filing, Proposed Tariff defines HILL as "A new commercial or industrial load, or increase in commercial or industrial load, at a single site connected through one or more shared Points of Interconnection (POIs) or delivery points, where such load is either (1) 10 MW or more if connected to the Transmission System at a voltage level less than or equal to 69 kV; or (2) 50 MW or more if connected to the Transmission System at a voltage level greater than 69 kV. An Electric Storage Resource is not considered a HILL. A load may be categorized as a HILL after the initial effective date of Attachment BA of the Tariff. High Impact Large Loads must register as such in accordance with Attachment AE of the Tariff and follow the processes specified in Attachment BA of the Tariff."

and (3) additional, on-going operational requirements for HILLs to ensure continued reliability of the Transmission System. In recognition of the unique opportunities presented by these loads, to SPP and nationally, SPP also proposed implementation of a new generator Interconnection Service[12] and related interconnection process to allow the limited connection of supporting generation for HILLs on an expedited basis, known as the High Impact Large Load Generation Assessment ("HILLGA").[13] A HILLGA Generating Facility will be studied in the same time frame as the HILL (90 days), which provides the HILLGA Generating Facility an expedited path to service to align with the HILL. Of note, the HILLGA process will not result in granting the Generating Facility traditional generator interconnection service, such as Network Resource Interconnection Service ("NRIS") or Energy Resource Interconnection Service ("ERIS"); instead, Generating Facilities studied under the HILLGA process will receive much more limited "Load Limited Resource Interconnection Service," or "LLRIS," which only studies the impact of the Generating Facility serving the identified HILL, which must be located in the same local area. If an interconnection customer with a Generating Facility supporting a HILL desires NRIS or ERIS, it will still be required to go through SPP's existing generator interconnection queue.

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[12] Tariff at Attachment V, Section 1 defines Interconnection Service as "the service provided by Transmission Provider associated with interconnecting Interconnection Customer's Generating Facility to the Transmission System and enabling it to receive electric energy and capacity from the Generating Facility at the Point of Interconnection, pursuant to the terms of the Generator Interconnection Agreement and, if applicable, the Tariff."

[13] SPP's High Impact Large Load Filing, Proposed Tariff defines HILLGA as "a planning assessment to evaluate the impact of new High Impact Large Load and supporting generation as described in Attachment BB of the Tariff."

As part of SPP's systematic approach in addressing the interconnection of large loads, SPP is currently developing two new Tariff processes to give additional opportunities for loads to connect to the SPP Transmission System. The first is a new conditional non-firm Transmission Service[14] process, known as Conditional High Impact Large Load Service ("CHILLS"). CHILLS will allow large loads the ability to quickly connect to the Transmission System without existing Designated Resources to support the large load or transmission requirements that may prevent the large load from connecting. Although this new Transmission Service process will allow the expedited connection of large loads, these large loads will be subject to curtailment or interruption until they have the requisite transmission and sufficient Designated Resources. However, when paired with SPP's proposed HILLGA process, Conditional HILL ("CHILL") customers may be able to hedge their curtailments by utilizing the HILLGA generation to support the CHILL. SPP is currently working with its stakeholders in the development of this new process.[15] It is anticipated that the SPP Board of Directors will vote on the new CHILLS transmission product in the February 2026 quarterly meeting.

The second new process to enable the connection of new loads is known as Price Adaptive Load Service ("PALS"). The PALS is a flexible, non-firm, Transmission Service option that will allow loads to vary their participation based on the price of energy. This new product is developed for any load, not just large loads, that is price sensitive and

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14 Tariff at Part I, Definitions T defines Transmission Service as "Point-To-Point Transmission Service provided under Part II of the Tariff on a firm and non-firm basis."

15 *See* Open Revision Requests posted at: https://www.spp.org/spp-documents-filings/?id=21069 (zip file for RR720 Conditional High Impact Large Load).

flexible (i.e., that does not need to be served from the Transmission System on a firm basis). PALS will allow these loads to participate in the SPP Integrated Marketplace through price sensitive demand bids in both the day-ahead and real-time markets. The PALS process is also under development in the SPP stakeholder process. It is anticipated that the SPP Board of Directors will vote on the PALS proposal in the May 2026 quarterly meeting.

III. COMMENTS

Please see below SPP's initial comments in response to the Proposed ANOPR. By providing initial comments, SPP does not waive its ability to comment on or object to issues in the future.

A. Large Load Thresholds

In the Proposed ANOPR, the Secretary of Energy states that, "the reforms should only apply to new loads greater than 20 MW and, for hybrid facilities, where the load is greater than 20 MW."[16] The Proposed ANOPR requests comments on this threshold of 20 MW.

In SPP's High Impact Large Load Filing, SPP proposed the following threshold to qualify as a HILL: "(1) 10 MW or more if connected to the Transmission System at a voltage level less than or equal to 69 kV; or (2) 50 MW or more if connected to the Transmission System at a voltage level greater than 69 kV." SPP proposed these two different thresholds for large loads to reflect the fact that the size of the load may impact the system differently when connected at different voltage levels. Specifically, a 10 MW HILL connecting to transmission facilities at or below 69 kV may have similar impacts to

[16] Proposed ANOPR at 10-11.

the Transmission System as a HILL that is 50 MW, or larger, connecting to transmission facilities that are greater than 69 kV.

SPP's proposed 50 MW HILL threshold was based primarily on information gathered by the North American Electric Reliability Corporation ("NERC") Large Load Task Force. The NERC Large Load Task Force requested surveys to determine what qualifies as a "large" quantity of load.[17] Most of the survey participants qualified "large" as greater than 50 MW. That survey also showed that "large" may be relative to other factors, such as voltage interconnection. As it relates to SPP's proposed 10 MW threshold, while the NERC definition of the Bulk Electric System includes only transmission elements of 100 kV or higher,[18] the additional 10 MW threshold is necessary due to the prevalence of 69 kV lines in the eastern interconnection portion of SPP's RTO footprint.

As demonstrated in SPP's High Impact Large Load Filing, the determination of a threshold for which loads are considered "large" is dependent upon the specific impacts these loads have on the system to which they are connected. SPP requests that any final rules in this matter do not include any specific MW threshold, but instead, the final rules should allow each transmission provider to propose thresholds specific to the needs of their transmission systems.

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[17] NERC, Characteristics and Risks of Emerging Large Loads, at 11 (July 2025), https://www.nerc.com/comm/RSTCReviewItems/3_Doc_White%20Paper%20Characteristics%20and%20Risks%20of%20Emerging%20Large%20Loads.pdf.

[18] NERC, Glossary of Terms Used in NERC Reliability Standards, (Oct. 1, 2025), https://www.nerc.com/pa/stand/glossary%20of%20terms/glossary_of_terms.pdf (definition of Bulk Electric System).

Additionally, system protection facilities may not fully protect the grid from stability issues, and as such, the full load at a hybrid facility should be considered when applying a MW threshold.

B. Additional Financial Commitments and Penalties

In the Proposed ANOPR, the Secretary of Energy suggests that "load and hybrid facilities should be subject to the standardized study deposits, readiness requirements, and withdrawal penalties."[19] The Proposed ANOPR requests comment on whether additional commitments or financial penalties would be appropriate.

In SPP's High Impact Large Load Filing, SPP proposed an application fee and increased deposits for requests to study and interconnect HILLs[20] and HILLGAs.[21] As stated in SPP's transmittal letter in its High Impact Large Load Filing, SPP asserts that these requirements for an application fee and increased deposits will ensure that highly speculative HILLS and HILLGAs are discouraged from applying to be studied.[22] As it relates to this Proposed ANOPR, SPP believes that any final rule related to the interconnection of large loads and supporting generation should include a level of financial commitment that ensures that SPP, and other transmission providers, are only studying those projects that fully intend to interconnect to their transmission system.

C. Study of Injection and/or Withdrawal Rights Requested

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19 Proposed ANOPR at 11.

20 SPP's High Impact Large Load Filing at Proposed Tariff, Attachment BA, Section 2.1.3.5.

21 SPP's High Impact Large Load Filing at Proposed Tariff, Attachment BB, Section 8.2.

22 SPP's High Impact Large Load Filing Transmittal letter at 22, 25, and 37.

The Proposed ANOPR states that "hybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested."[23] The Proposed ANOPR goes on to state, as an example, "[f]or example, a hybrid facility consisting of a 500 MW load and a 600 MW generating facility may seek no withdrawal rights and 100 MW of injection rights."

SPP believes it is important from a study perspective to ensure that any injection or withdrawal of MW to/from the Transmission System appropriately accounts for all impacts to the Transmission System to ensure the reliable operation of the grid. Using the Proposed ANOPR's example above, if a 500 MW load is always connected to the Transmission System, even if not always taking the full MW quantity of withdrawals necessary to serve the load, SPP cannot appropriately study the actual impacts of that load on the Transmission System without studying the full impact of its potential withdrawals from the Transmission System. Although this load may have a generator capable of offsetting the load at most times, in the event that generator goes offline, SPP would have to provide the full MW amount to serve that load, and would, therefore, experience the effects of its full withdrawal even if only temporarily. If SPP does not study the impact of the entirety of the load, then SPP would not know the impact the load would have when the generator is not providing the full generation necessary to cover the 500 MW load.

Even with system protections in place, very rapid withdrawals of large amounts of power can adversely affect the transmission grid (e.g., through rapid frequency excursions and unintended protection operations). All generators are susceptible to unexpected tripping, and protection schemes themselves may fail under fast dynamic conditions. This

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[23] Proposed ANOPR at 11.

risk is acknowledged in Institute of Electrical and Electronics Engineers (IEEE) Standard 2800-2022, Clauses 7.3 (*Frequency Ride-Through*) and 9.2 (*[rate of change of frequency] Protection*),[24] which require inverter-based resources to maintain ride-through capability and ensure proper coordination of rate of change of frequency protection with system operator settings. If SPP does not study the impact of the entirety of the load, SPP would not know what impact the load would have if the generator doesn't provide the 600 MW of generation.

As it relates to the studying of the injection of the hybrid facility, SPP would also need to study the entire injection of that Generating Facility[25] to ensure that all impacts are captured due to the injection. If the hybrid facility mentioned in the Proposed ANOPR example will never inject more than 100 MW into the SPP Transmission System, SPP could study that and limit the injection rights to 100 MW. But if the hybrid facility is injecting the entirety of its 600 MW injection onto the SPP Transmission System, SPP must have the ability to study this full impact. If SPP is required to only study the net impact of this injection, SPP would not know how much impact this Resource would have on the

[24] Institute of Electrical Engineers, IEEE Standard for Interconnection and Interoperability of Inverter-Based Resources (IBRs) Interconnecting with Associated Transmission Electric Power Systems (IEEE 2800-2022).

[25] Tariff at Attachment V, Section 1 defines Generating Facility as **"**Interconnection Customer's device(s) for the production and/or storage for later injection of electricity identified in the Interconnection Request, but shall not include Interconnection Customer's Interconnection Facilities and shall not include a Storage as Transmission Only Asset as defined in Section 1 of the Tariff. A Generating Facility consists of one or more generating unit(s) and/or storage device(s) which usually can operate independently and be brought online or taken offline individually."

SPP Transmission System at times when that Resource is injecting 600 MW and the load is taking 0 MW.

SPP is not specifically opposed to the development of new transmission service products that could be used to serve large loads with or without hybrid facilities, but if the Commission requires changes to SPP's current long-term transmission service studies and process through these ANOPR proceedings, the Commission should ensure that the Transmission System is fully protected from any Resource's possible injections/withdrawals or any load's withdrawals if they are connected to the SPP Transmission System. To ensure reliability and adherence with NERC standards, SPP must consider the possible impact of all loads, including contingencies of equipment and system stability. System stability concerns exist even if net power flow could be guaranteed.

D. Transmission Service Charges

The Secretary of Energy makes statements in the Proposed ANOPR that relate to how transmission charges should be billed based on the energy being withdrawn from the Transmission System. First, the Proposed ANOPR states "utilities serving large loads, including those at hybrid facilities, should be responsible for transmission service based on their withdrawal rights, as that value amount reflects the quantity of capacity and energy that is being transmitted across the transmission system to the load."[26] Also, the Proposed ANOPR states "utilities serving large loads, including those at hybrid facilities, should be responsible for ancillary services based on peak demand, without consideration of any co-

[26] Proposed ANOPR at 13.

located generation. Any co-located generating facilities will similarly be fully compensated for the provision of ancillary services."[27]

If the Commission mandates a transmission cost netting paradigm, which is implied by the hybrid example, consideration should also be given to the converse scenario, where a facility with 600 MW of load and 500 MW of generation has 100 MW of withdrawal rights. In this case, billing for transmission based upon coincident peak usage can be particularly problematic if the hybrid resource requires transmission to be built and maintained, yet the hybrid facility avoids the coincident peak upon which billing is currently calculated. While billing on energy usage is a possible alternative, the Commission may also consider billing strictly based upon the reserved amount, regardless of usage, to avoid undue cost shifts to other consumers.

The SPP Tariff currently allows for eligible entities to either take Network Integration Transmission Service ("NITS") or Point-To-Point ("PTP") Transmission Service to serve load connected to the SPP Transmission System. NITS is billed based on the Network Customer's coincident peak load. PTP Transmission Service is billed to the transmission customer on the reserved MW capacity. These transmission charges are appropriate based on how SPP designs and constructs the SPP Transmission System. Both NITS and PTP Transmission Service are firm Transmission Service products and, as such, SPP constructs its Transmission System to serve this load on a continuous and reliable basis. Currently, SPP does not have a Transmission Service product that charges customers based on the exact amount of energy that is withdrawn from the system.

[27] Proposed ANOPR at 13.

As discussed above, SPP and its stakeholders are currently developing the PALS, which will allow a flexible, non-firm, Transmission Service option that will allow loads to vary their participation based on the price of energy. The pricing of this service is still being determined, but this pricing must ensure that Transmission Owners[28] are being wholly reimbursed for the usage of their transmission facilities.

A usage-based transmission charge is a fundamental change in how SPP currently charges for Transmission Service across the SPP Transmission System. If the Commission determines in a final rule that large load customers should only be charged for transmission service based on their withdrawal rights, SPP urges the Commission to consider all costs incurred by granting this type of transmission service to ensure that costs are allocated appropriately so that transmission owners are held harmless by the usage.

E. Cost Responsibility of Interconnection Request

The Proposed ANOPR proposes that: "load and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies. We seek comment on whether such costs should be offset through a crediting mechanism and, if so, over how many years."[29]

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28 Tariff at Part I, Definitions T defines Transmission Owner as "each Member of SPP that has executed an SPP Membership Agreement as a Transmission Owner and therefore has the obligation to construct, own, operate, and maintain transmission facilities as directed by the Transmission Provider and: (i) whose Tariff facilities (in whole or in part) make up the Transmission System; or (ii) who has accepted a Notification to Construct but does not yet own transmission facilities under SPP's functional control. Those Transmission Owners that are not regulated by the Commission shall not become subject to Commission regulation by virtue of their status as Transmission Owners under this Tariff; provided, however, that service over their facilities classified as transmission and covered by the Tariff shall be subject to Commission regulation."

29 Proposed ANOPR at 12.

It should be noted here that the Commission-approved SPP Bylaws designate specific authorities within the SPP governance to the SPP Regional State Committee[30] ("RSC").[31] Specifically, as it relates to cost responsibility of network upgrades in SPP, Section 7.2 of the SPP Bylaws delegate primary responsibility for determining regional proposals for cost allocation to the RSC. Consistent with the Commission-approved SPP Bylaws, SPP requests that any final rule adopted based on this Proposed ANOPR provide the RSC with the opportunity to provide significant input into how network upgrade costs will be allocated within SPP as it relates to the interconnection of large loads.

SPP currently has provisions in its Tariff to credit eligible entities who pay for construction of upgrades on the SPP Transmission System.[32] Entities who qualify under Attachment Z2 of the Tariff are compensated for eligible upgrade costs through Incremental Long Term Congestion Rights ("ILTCRs"). ILTCRs are limited to at least ten years and no more than twenty years.[33] SPP requests that any crediting mechanism allowed under any proposed rule in this docket be consistent with Attachment Z2 and Attachment J of the SPP Tariff.

F. Curtailable Load

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[30] The SPP Regional State Committee provides collective state regulatory agency input on matters of regional importance related to the development and operation of bulk electric transmission. The SPP RSC is comprised of retail regulatory commissioners from agencies in Arkansas, Iowa, Kansas, Louisiana, Minnesota, Missouri, Nebraska, New Mexico, North Dakota, Oklahoma, South Dakota and Texas.

[31] *See* Southwest Power Pool, Inc., Governing Documents Tariff, Bylaws, Section 7.2.

[32] *See* Tariff at Attachment Z2.

[33] *See* Tariff at Attachment J, Section V, A-C.

The Proposed ANOPR states: "[t]he interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited. The system operator's ability to control such facilities through curtailment and/or dispatch must be sufficient for the system operator to integrate the facility into both operations and system planning. This ensures the timely and orderly addition of large loads to the transmission system in a safe, reliable, and non-discriminatory manner. We seek comment on whether this should be accomplished through a serial interconnection study process or by some other means. We also seek comment on appropriate deadlines for such an expedited study process, including whether such studies can be completed in 60 days."[34]

As stated above, SPP and its stakeholders are currently developing a new conditional transmission service product (i.e., CHILLS) that would allow customers to take Transmission Service for their HILL, even if they do not have sufficient Designated Resources to serve the load or there are transmission constraints prohibiting the full interconnection, but this customer must be willing to have their load curtailed or interrupted based upon system conditions. SPP has received numerous concerns from stakeholders related to the current proposal, such as: (1) market price impacts; (2) Transmission System reliability; (3) impact to emissions permit-limited Resources, such as combustion turbines, and (4) the frequency and duration of curtailments.

Based on SPP's experience, 60 days is not sufficient time to study the reliability impacts of large loads. SPP's new HILL policy in the High Impact Large Load Filing provides for a 90 day study time to allow for the necessary stability studies, such as

[34] Proposed ANOPR at 12.

transient stability analysis, short-circuit ratio analysis, and a critical clearing time screening to determine if further electromagnetic transient stability (EMS) analysis is needed. These large loads—potentially exceeding hundreds of megawatts—pose substantial operational and planning challenges for all transmission service providers in the United States.

In NERC's incident review relating to recent events in the Eastern Interconnection,[35] NERC highlighted that rapidly growing voltage-sensitive large loads, such as data centers and cryptocurrency facilities, can cause significant load loss during normal fault clearing events, posing emerging reliability risks for the bulk electric system. Also, in a recent NERC and Regional Entity analysis,[36] they identified multiple bulk electric system disturbances in 2024–2025 where over 1,000 MW of unexpected large load reductions occurred. These events highlight growing reliability risks associated with the rapid increase and unpredictable behavior of large loads across the grid.

On April 28, 2025, a blackout in the Spanish Peninsular Electrical System underscored the impact of inverter-based resources and large industrial loads on system reliability and operational dynamics.[37] Without a defined process for their integration and operation, such loads can create local voltage instability, thermal overloads, and frequency

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35 NERC Incident Review, Considering Simultaneous Voltage-Sensitive Load Reductions, https://www.nerc.com/globalassets/our-work/reports/event-reports/incident_review_large_load_loss.pdf(Jan. 8, 2025).

36 NERC Alert, Industry Recommendation: Large Load Interconnection, Study, Commissioning, and Operations, https://www.nerc.com/globalassets/programs/bpsa/alerts/2025/nerc-alert-level-2--large-loads.pdf (Sep. 9, 2025).

37 Red Eléctrica, Blackout in the Spanish Peninsular Electrical System the 28th of April 2025, https://d1n1o4zeyfu21r.cloudfront.net/WEB_Incident_%2028A_SpanishPeninsularElectricalSystem_18june25.pdf (June 18, 2025).

disturbances, particularly under contingency or restoration conditions. Establishing a structured framework ensures consistent technical evaluation, reliable interconnection studies, and transparent market participation requirements.

Planning risks for the interconnection of HILLs include accelerated construction timelines, supply chain constraints, co-location with generation resources, adequacy of generation capacity and energy, limitations in transmission planning studies, multi-sector electrification, and aggregate load growth. To mitigate these risks, SPP proposed employment of several proactive strategies. In SPP's High Impact Large Load Filing, SPP proposed to utilize an accelerated 90 day study process, including increased study deposits, for large load additions that successfully pass short-circuit ratio, critical clearing time screening, and did not have issues identified in transient stability analysis. SPP is also exploring flexible interconnection approaches to address supply chain delays, including the potential deployment of grid-enhancing technologies.

There are significant operational risks that encompass the interconnection of large loads, including: variability and fluctuations, cybersecurity vulnerabilities, price-sensitive large loads, ride-through performance, power quality or bulk electric system performance issues, and maintenance and outage coordination. SPP proposed several requirements in its High Impact Large Load Filing to mitigate these identified risks. SPP proposed to require non-conforming load forecasts, ramp rate restrictions, installation of phasor measurement recording and communication equipment, and compliance with ride-through performance standards.[38] As part of the High Impact Large Load Filing, SPP developed a Ride-Through

[38] *See* SPP's High Impact Large Load Filing at 18-21.

Requirements Manual to establish voltage and frequency ride-through requirements.[39] Additionally, SPP may require power quality studies as part of the supplemental studies outlined in the approved HILL Delivery Point Studies (HDPS) Manual.[40] Lastly, SPP also is developing proposed requirements for large loads to submit outage and derate information and coordinate maintenance activities in accordance with its outage coordination policy.

Any load connected to the SPP Transmission System, including load that agrees to be curtailed, will require energy to be produced and will take some capacity from the Transmission System; otherwise, they would not need to connect to the grid. Even when agreeing to curtail, a curtailable, or non-firm load, will impact existing and future firm load and should not be prioritized over a load that is willing to bring, or already has Designated Resources. If anything, loads with sufficient resources should be prioritized.

Serial interconnection studies are preferred if the volume of requests is relatively low and one request does not impact another. However, cluster studies may be needed if volumes become large to mitigate the time it may take to perform the studies, and potential restudies in a serial manner. Cluster studies will take longer than 90 days, and the impacts of customers withdrawing will need to be sufficiently mitigated to avoid the issues of backlog that have occurred with generation interconnection queue processes. SPP suggests allowing each region to determine the best approach to minimize the time of study.

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39 The High Impact Large Load Ride-Through Requirements were approved through the SPP stakeholder process by the necessary working groups. *See* Operating Reliability Working Group Meeting Materials, dated October 8, 2025, at Agenda Item 7, posted at: https://www.spp.org/spp-documents-filings/?id=19845.

40 *See* SPP's High Impact Large Load Filing at Proposed Tariff, Attachment BB, Sections 2 and 3.

As the Commission develops its final rules relative to this Proposed ANOPR, it is imperative that it considers the concerns outlined above. If a large load is allowed to interconnect to the Transmission System on a long-term firm basis, there must be corresponding availability of generation capacity to serve the load addition. Even if loads do not take service at peak times, there must still be energy to serve the load, which depends on the instantaneous level of all other loads and the availability of all other Resources. The availability of Resources is dependent upon numerous, varied factors, such as, but not limited to, fuel availability, state of charge, demand response hours available, permit limits, and outages. Any "non-firm" loads must only be allowed to take service when these factors would not significantly impact SPP's ability to serve firm load at both the time service is taken, as well as over any potential time period of impact. For permit limitations, time periods of impact may be different depending on the Resource. NERC has recognized this concern, at least in part, with newly proposed standards on energy assurance.[41] Without sufficient generation capacity, and energy to serve all new load additions along with the existing load, there will be detrimental impacts to the transmission grid, even if those loads are curtailable or interruptible.

G. Option to Build for Load Interconnection Customers

The Proposed ANOPR proposes "to the extent the interconnection customer is not the transmission owner, the interconnection customer shall be afforded the same (or equivalent) option to build as currently provided to generator interconnection customers."[42]

[41] *See* NERC Project 2024-02, Planning Energy Assurance.

[42] Proposed ANOPR at 13.

In considering whether to implement this proposed requirement, the Commission should be mindful of state laws that restrict who can build and own transmission. Network Upgrades assigned through SPP's transmission service process can only be assigned to entities that request transmission service. If an entity is not eligible to become an SPP Transmission Customer because it is not eligible to serve load in a state in the SPP region, that entity can only build a transmission asset in SPP, pursuant to the Commission-approved Tariff, if that entity agrees to sponsor the upgrade and pay for the entirety of the costs.

IV. CONCLUSION

SPP respectfully requests that the Commission consider these comments in developing any final rules that result from the Proposed ANOPR.

Respectfully submitted,

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